Sparse Routing Improves Low-Resource Translation

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Abstract

We introduce SparseRoute, a mixture-of-experts layer that routes each token to a single expert. On the FLORES-200 benchmark SparseRoute improves BLEU by 2.1 points over a dense Transformer baseline while using 30% fewer training FLOPs. Code is available at https://github.com/example/sparseroute.

1 Introduction

Recent work has shown that careful design of the training objective matters as much as model size. We study this question under a controlled compute budget and report results averaged over several random seeds so that differences can be attributed to the method rather than to noise. Recent work has shown that careful design of the training objective matters as much as model size. We study this question under a controlled compute budget and report results averaged over several random seeds so that differences can be attributed to the method rather than to noise. Recent work has shown that careful design of the training objective matters as much as model size. We study this question under a controlled compute budget and report results averaged over several random seeds so that differences can be attributed to the method rather than to noise.

Our contributions are threefold. First, we propose a routing rule that needs no auxiliary load-balancing loss. Second, we evaluate it on FLORES-200 and WMT 2014 English-German. Third, we release code and trained checkpoints.

2 Related Work

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3 Method

Each token representation x is sent to the expert with the highest router score, and the layer output is the expert output scaled by the router probability:

$$y = p_k(x)\, E_k(x), \quad k = \mathrm{argmax}_i\, W_r\, x \qquad (1)$$

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than to noise.

3.1 Training details

We train with Adam using a learning rate of 3e-4, batch size 256 and 50k steps on 8 NVIDIA A100 GPUs, which takes about 36 hours. Dropout is 0.1. We use 3 random seeds.

4 Experiments

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Method	FLORES-200 BLEU	WMT14 En-De BLEU	FLOPs (%)
Transformer	24.3 ± 0.2	27.1 ± 0.1	100
Switch	25.0 ± 0.3	27.4 ± 0.2	72
SparseRoute (ours)	**26.4 ± 0.2**	**27.9 ± 0.1**	70

Table 1: Test BLEU averaged over 3 seeds (mean ± std).

4.1 Ablation study

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5 Limitations

We only evaluate translation. Routing may behave differently on long documents.

6 Conclusion

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